

Maxime Baelde

Ph. D, Data Scientist

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Work experiences

March 2024 – **Senior Data Scientist**, ANTVOICE, Paris.

Today I own the machine learning behind AntVoice's real-time bidding platform and its LLM-agent products for advertisers.

- Design and own the platform's core ML models for real-time bidding, win prediction, and click/conversion forecasting, including a reinforcement-learning bidding agent
- Build and evaluate LLM multi-agent systems for advertisers: evaluation frameworks, agentic security, and automatic prompt optimization
- Industrialize data pipelines and MLOps: Airflow on GKE, BigQuery, CI/CD
- Tech: Python, GCP, BigQuery, Airflow, Docker

June 2026 – **Founder**, DUETLY, Lille.

Today Designed and built solo a SaaS that streamlines file transfer and revision management between audio engineers and artists, replacing WeTransfer, WhatsApp, and email.

- Audio analysis engine (LUFS, True Peak, LRA), billing infrastructure, real-time messaging, bilingual (FR/EN) interface
- File delivery, in-browser waveform playback, time-coded annotations, versioned revision cycles, quotes & invoices
- Tech: Next.js, React, TypeScript, Firebase, Stripe
- Live at duetly.app

May 2022 – **Data Scientist Consultant**, CONSORTIA, Lille.

February Context: Retail

- 2024
- Develop and maintain sales forecast algorithms
 - Backtest and implementation of new algorithm to forecast daily store sales
 - Adaptation of existing DAGs to run the tasks on Airflow
 - Techno: GCP, BigQuery, GCS, AI Notebook, Airflow, Github, git
- Participate in the recruitment of Data Scientists, Data Analysts and Data Engineers

October 2019 **Lecturer in Statistics**, ENIGMA SCHOOL, Lille.

– Today Lectures of Statistics with R and Python at Enigma School (Master in Computer Science, Big Data)

July 2022 – **Mixing/Mastering Engineer**, FREELANCE, Lille.

Today Produce, record, mix and master songs for the bands The Pyloons and Fracton.

July 2019 – **Machine Learning Engineer, Data Scientist**, WAVELY, Villeneuve d'Ascq.

- April 2022
- Development of Python packages about signal processing and machine learning.
 - Experiment deep learning methods for embedded audio source classification, using PyTorch.
 - Conversion to ONNX format to run the algorithms on small computers
 - Dockerization of audio classification service
 - Update SQLite database schemas using SQLAlchemy
 - Techno: Git, Python, SQL, Docker, Ansible, Grafana, Linux

- November 2017 – **Part-time lecturer in Signal Processing**, ECOLE CENTRALE DE LILLE, Villeneuve d'Ascq.
- September 2022 Involved in lectures of Signal Processing at Ecole Centrale de Lille (Bachelor and Master equivalent lectures)
- October 2015 – July 2019 **R&D Engineer, Audio / Machine Learning**, NAHIMIC – A BRAND OF STEELSERIES, Villeneuve d'Ascq.
Develop algorithms related to probabilities, statistics, machine learning and artificial intelligence, for applications to digital audio technologies. Main research topics: audio source classification, audio source separation, generative models, deep learning.
- January 2016 – December 2018 **Ph. D Student**, A-VOLUTE – INRIA, Villeneuve d'Ascq.
Thesis subject: Real-time identification, localization and separation of audio sources in multichannel audio streams. Supervisor: Christophe Biernacki, Professor (Université de Lille, INRIA, CNRS)
- September 2018 – October 2018 **Part-time lecturer in Statistics**, ISA, Lille.
Involved in lectures of Statistics with R at ISA (Bachelor equivalent lectures)
- April 2015 – September 2015 **Internship – R&D Engineer**, A-VOLUTE, Roubaix.
Modelization of HRTF filters and optimization for 3D audio effect.
 - Implementation of Machine Learning techniques (regression) for HRTF individualization
 - Implementation of a non rigid registration algorithm for 3D scans
- May 2014 – August 2014 **Internship – Research assistant**, UNIVERSITÉ LIBRE DE BRUXELLES, Brussels.
Development of an experimental method for assessing the musical qualities of violin strings. Supervisor: Jean-Pierre Herman

Computer skills

Software programming	GIT, GITHUB, PYTHON (NUMPY, PANDAS, ...), R, BASH, C/C++
Web development	TYPESCRIPT, NEXT.JS, REACT.JS, FIREBASE, STRIPE
Machine Learning	SCIKIT-LEARN, PYTORCH
Data Visualization	GOOGLE DATA STUDIO, TABLEAU, POWER BI, GRAFANA
Data Engineering	GOOGLE BIGQUERY, SQLALCHEMY, SQLITE, POSTGRESQL
MLOps	AIRFLOW, DOCKER, MLFLOW, DVC, CI/CD
Audio	FFMPEG, CUBASE, WAVELAB PRO
Operating system	WINDOWS, LINUX

Open-source projects

- churn-poc End-to-end churn prediction project: model training to API exposition with FastAPI. [GitHub]

generative-audio-source-models	Generative models for real-time audio source classification and separation (PhD thesis, Univ. Lille 2019). [GitHub]
nongaussian-mixtures	Scikit-learn-compatible mixture models (Dirichlet, Beta, binned Gaussian) from my thesis appendices. [GitHub], [PyPI]
simple-vad	Voice Activity Detector: signal-processing beamforming + CNN classifier. [GitHub]
sober-selfhost	Energy-sober self-hosting on consumer hardware: on-demand containers (Sablier) gated behind Caddy, multi-DB encrypted backups (Postgres/MariaDB/SQLite -> rclone crypt), bidirectional Wake-on-LAN over Tailscale, and battle-tested Docker healthcheck fixes. [GitHub]

Scientific communications

- 2026 **Journal article (under review): TMLR.**
Maxime Baelde. “Threshold-Cascaded Lexicographic Rewards for Multi-Constrained RL in Real-Time Bidding”. Submitted to TMLR (Transactions on Machine Learning Research), under review. [openreview]
- 2021 **Patent.**
Maxime Baelde, Laurent Mareuge and Julien Roland. “Système de reconnaissance et d’identification de sources sonores en temps réel”. FR Patent FR3109458B1. [patent]
- 2020 **Patent.**
Maxime Baelde, Nathan Souviraà-Labastie and Raphaël Greff. “Apparatus and method for audio analysis”. US Patent US11956616B2. [patent]
- 2019 **Ph.D thesis.**
Maxime Baelde. “Modèles génératifs pour la classification et la séparation de sources sonores en temps-réel”. Ph.D thesis, Université de Lille, 2019. [thesis], [code]
- 2019 **Journal article: Pattern Recognition.**
Maxime Baelde, Christophe Biernacki and Raphaël Greff. “Real-Time monophonic and polyphonic audio classification from power spectra”. In : Pattern Recognition 92 (august 2019), p. 82-92. [paper]
- 2019 **Pre-print: Grets 2019.**
Nathan Souviraà-Labastie, **Maxime Baelde**, Thomas Malet and Raphaël Greff. “Impact des conditions d’attaques sur les contre-mesures pour la reconnaissance du locuteur”. pre-print Grets. [paper]
- 2019 **Pre-print paper: Interspeech 2019.**
Maxime Baelde, Nathan Souviraà-Labastie and Raphaël Greff. “Influence of the attack conditions on countermeasures for Automatic Speaker Verification”. pre-print Interspeech. [paper]
- 2017 **Conference paper: ICASSP 2017.**
Maxime Baelde, Christophe Biernacki and Raphaël Greff. “A mixture model-based real-time audio sources classification method”. 2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 5-9 Mar 2017, pp. 2427-2431, New Orleans, USA. [paper], [poster]
- 2017 **Conference paper: JDS 2017.**
Maxime Baelde, Christophe Biernacki, Raphaël Greff. “Classification de signaux audio en temps-réel par un modèle de mélanges d’histogrammes”. 49èmes Journées de Statistiques, June 2017, Avignon, France. [paper]

- 2015 **Master Thesis.**
Maxime Baelde. "Modelization of HRTF filters and optimization for 3D audio effect".
 Master thesis, 2015, Université de Lille. [paper]
- 2014 **Abstract: JASA 2014.**
Maxime Baelde, Jessica De Saedeleer et Jean-Pierre Hermand. "Experiment to evaluate musical qualities of violin strings". In : The Journal of the Acoustical Society of America 136.4 (2014), p. 2284-2285. [abstract]

Education

- 2023 **Training, DJ Network, MIXING & MASTERING** (expert module): Mixing and processing audio tracks of a musical production.
 RNCP35223BC04 block - Mixing and processing audio tracks of a musical production
- 2016–2019 **Ph.D thesis,** *Université de Lille*, Villeneuve d'Ascq, Modèles génératifs pour la classification et la séparation de sources sonores en temps réel.
- 2012–2015 **Engineering Degree,** *École Centrale de Lille*, Villeneuve d'Ascq, *Decision making and Data analysis.*
- 2014–2015 **Master Research in Applied Mathematics,** *Université de Lille*, Villeneuve d'Ascq.
- 2010–2012 **Preparatory classes: two-year undergraduate intensive course in mathematics and physics,** *Lycée Henri Wallon*, Valenciennes.
- 2007–2010 **A-Level, engineering sciences and physics speciality,** *Lycée Pierre Forest*, Maubeuge.

Language

- French **Mother Tongue**
- English **Technical and professional**

Hobbies

- Music (technical) Compose, produce, record, mix and master songs
- Music (artistic) Play the piano (since 19 years) and the guitar (since 22 years)
- Videogames RPG, Tactical, Action
- Reading Thriller (Maxime Chattam), science-fiction (Isaac Asimov, Alain Damasio,...)
- Others Bakery, Gardening and food autonomy